

Supplementary Winter Examination – 2024

Course: B.TECH Branch : Electrical Engineering/Electrical Engg(Electronics & Power)/
Electrical & Electronics Engg/ Electrical & power Engineering

Subject Code & Name: BTEEC703 Electrical Drives

Semester : VII

Max Marks: 60

Date:10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per CBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculator is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(CO)	Mark
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is NOT a typical part of an electrical drive?	CO1	
	a) Power modulator b) Electric motor c) Control unit d) Heat exchanger		
	Which of the following is a multiquadrant operation?	CO1	
	a) Forward motoring b) Forward braking c) Reverse motoring d) All of the above		
	Load equalization in drives is achieved by:	CO2	
	a) Increasing motor power b) Using flywheels c) Improving insulation d) Increasing supply voltage		
4	In steady-state operation, the drive operates under conditions of:	CO2	
	a) Constant speed and torque b) Varying acceleration c) Maximum load variations d) Zero feedback		
	Current limit control in drives ensures that:	CO2	
	a) The drive operates at maximum efficiency b) Motor current does not exceed safe limits c) Speed remains constant d) Feedback is minimized		
5	Static Kramer drives are typically used for:	CO3	
	a) Low power applications b) Controlling the speed of slip ring c) Single-phase motors d) Enhancing motor braking		

		induction motors				
7	Which type of DC motor drive is most suitable for dynamic braking?				CO3	
	a) Single-phase full converter drive	b) Dual converter drive	c) DC-DC converter drive	d) Half-wave drive		
8	Multiquadrant operation in induction motor drives requires:				CO4	
	a) Stator voltage control	b) Voltage source inverters	c) Rotor resistance variation	d) Constant torque		
9	Load commutated inverters are typically used for:				CO4	
	a) Starting synchronous motors	b) Speed control of synchronous motors	c) Reducing load fluctuations	d) Braking applications		
10	Static variable frequency control for synchronous motors is primarily used for:				CO4	
	a) High-speed operations	b) Dynamic braking	c) Energy-efficient operations	d) Constant frequency applications		
11	In textile mills, drives must primarily focus on:				CO5	
	a) High torque at low speeds	b) Precise speed control and torque control	c) Reducing power losses	d) Operating at fixed speeds		
12	Which type of motor is typically used for kiln drives in cement mills?				CO5	
	a) Synchronous motor	b) Slip ring induction motor	c) Permanent magnet motor	d) Single-phase AC motor		
Q.2	Solve the following.					12
A)	Draw block diagram of electrical drives and explain its different parts.				CO1	6
B)	Explain four quadrant operation of motor driving hoist load.				CO1	6
Q.3	Solve the following.					12
A)	Write a brief note on classes of duty for on electric motor in on electric drive				CO2	6

B)	Describe with neat block diagram of Closed Loop Speed Control of Multi Motor Drives:	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	A 200 V, 875 rpm, 150 A separately excited dc motor has an armature resistance of 0.06Ω . It is fed from a single phase fully-controlled rectifier with an ac source voltage of 220 V, 50 Hz. Assuming continuous conduction, calculate (1) firing angle for rated motor torque and 750 rpm	CO3	6
B)	Explain braking technique of DC motor in detail : 1) Plugging 2) Regenerative braking	CO3	6
C)	Explain the working and characteristics of a single-phase full converter drive for controlling the speed of a DC motor.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain operation of VSI fed induction motor drive.	CO4	6
B)	Discuss the Static Scherbius Drive for slip power recovery in induction motors. Explain its working with a block diagram.	CO4	6
C)	Explain the concept of voltage and frequency control for induction motor drives	CO4	6
Q.6	Solve Any Two of the following.		12
A)	Explain in brief drives required for various stages in textile mills	CO5	6
B)	Draw the block diagram of variable frequency control of multiple synchronous motor drive and explain in brief.	CO5	6
C)	Explain the principle of static variable frequency control for synchronous motors. How does it affect speed and torque control?	CO5	6
*** End ***			

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